



PadhAI

Offline On-Device AI Study Companion

Snapdragon® AI Lab Build & Present Challenge — Qualcomm

Submitted by Ayush · Individual Participation

The Problem

AI tutoring today assumes connectivity, cost, and comfort with sharing your notes

No internet, no AI

Students in hostels, tier-2/3 towns, and low-bandwidth homes can't reliably use cloud AI tutors.

Cost per query

Cloud LLM tools charge per use or require subscriptions — a real barrier for students.

Privacy trade-off

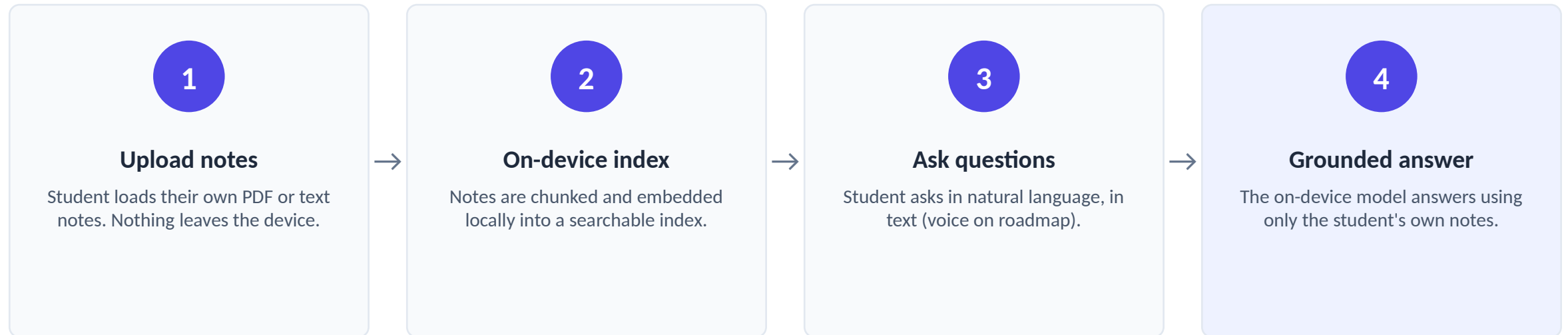
Personal notes and questions are sent to remote servers with every query.

Offline tools are static

Existing offline study apps are flashcard decks and PDFs — no real intelligence.

The Solution

A private AI tutor that runs entirely on-device — no internet required



Also auto-generates revision flashcards from the same notes.

Architecture — Powered by Qualcomm AI Hub

Every stage runs on the Snapdragon Hexagon NPU via ONNX Runtime's QNN Execution Provider

Pipeline Stage	Qualcomm AI Hub Model	Runs On
Note embeddings (RAG)	MiniLM-L6-v2 (quantized)	Hexagon NPU
Answer generation	Llama-3.2-3B-Instruct (INT4/8)	Hexagon NPU
Voice question input	Whisper-Base-En	Hexagon NPU
Scanned-PDF OCR	PaddleOCR / EasyOCR	Hexagon NPU

100% local inference — zero network calls anywhere in the pipeline.

Technical Implementation

A working demo today, with a clean, documented path to Snapdragon NPU deployment

Demo Engine (runs now)

- Streamlit app, runs on any laptop instantly
- TF-IDF + cosine similarity retrieval (scikit-learn)
- Extractive answers from the student's own notes
- Heuristic flashcard generator
- Zero downloads, zero GPU/NPU required to try

NPU Production Engine (Snapdragon)

- onnxruntime-qnn + QNNExecutionProvider
- MiniLM-L6-v2 embeddings on Hexagon NPU
- Llama-3.2-3B-Instruct for generative answers
- Whisper-Base-En for voice questions
- Single documented swap point: NPUIferenceEngine

Deployment, Accessibility & Impact

Built for the realities of Indian students' connectivity and budgets

Zero connectivity needed

Works fully offline after setup — hostels, villages, flights.

Zero recurring cost

No API tokens or subscriptions, unlike cloud AI tutors.

Privacy by design

Notes and questions never leave the device.

No discrete GPU needed

Quantized models fit the Hexagon NPU's power/thermal envelope.

Roadmap & Closing

From working demo to a Snapdragon-native product

- Activate real AI Hub ONNX models on a Snapdragon X-series HP device; benchmark NPU latency/power
- Add Whisper-based voice Q&A
- Multi-subject notebooks with per-subject indexes
- Package as a signed native Windows app (no dev setup needed)
- Pilot with a college class for real usage data

PadhAI makes on-device AI personal, private, and accessible — turning every Snapdragon-powered HP PC into a student's own offline tutor.